

“Intelligent Water Distribution Monitoring and Leakage Detection Platform”

CO5 ASSESSMENT TOOL 1 - RESEARCH PAPER REVIEW / LITERATURE SURVEY

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DECLARATION

I, Lenin K, of the Department of Computer Science and Engineering, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the assignment work entitled “Intelligent Water Distribution Monitoring and Leakage Detection Platform - Research Paper Review / Literature Survey” is the result of my own bonafide effort. All research papers reviewed in this document are cited to their original authors and publication venues, and to the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity.

Place: Chennai

Date: 22/08/2026

Signature of the Student

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1. Introduction

1.1 Problem Statement and Scope of the Survey

The assigned problem statement is an Intelligent Water Distribution Monitoring and Leakage Detection Platform: an IoT-enabled system that continuously monitors flow, pressure and water-quality telemetry across a municipal pipeline network, applies a leak-detection algorithm within defined District Metering Areas (DMAs), raises alerts, automates maintenance work orders, and allows authorized operators to remotely actuate valves and pumps through a SCADA-integrated control layer. Before designing or refining such a system, it is necessary to understand how the research community has approached the same underlying problem: detecting and localizing leaks in a water distribution network using sensing, communication and analytics technologies.

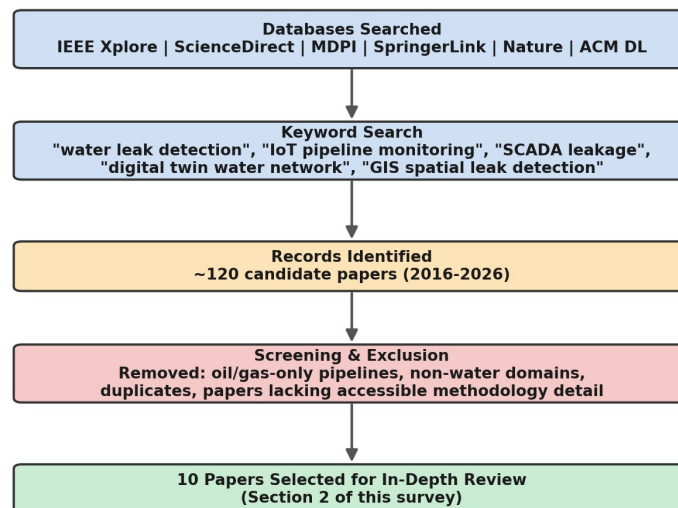
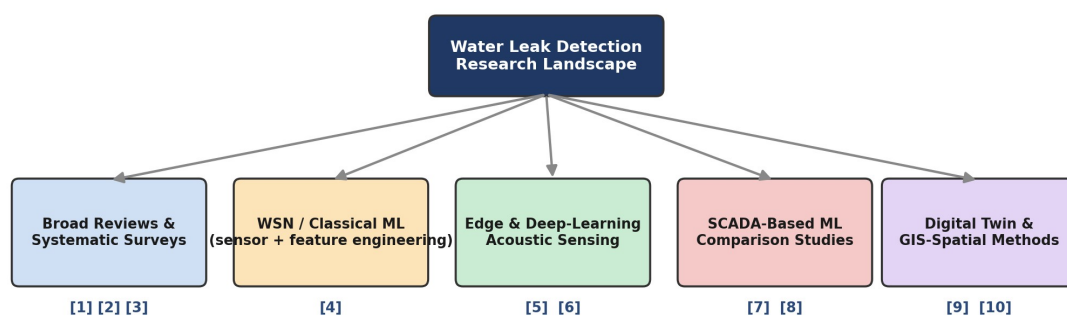
This survey reviews ten peer-reviewed research papers and systematic reviews (2019-2026) that address water-pipeline leak detection using Internet of Things (IoT) sensing, wireless sensor networks, machine learning and deep learning, SCADA-integrated analytics, digital twins, and GIS-based spatial methods. Together these papers span the sensing layer, the edge/cloud analytics layer, and the operational/deployment layer that the assigned platform must also address.

1.2 Objectives of the Literature Survey

- To identify credible, relevant, recent research papers addressing IoT-based, ML/DL-based, SCADA-based, digital-twin-based, or GIS-based water leak detection.
- To summarize each paper's objective, methodology, technology stack, dataset, and key findings in a consistent format.
- To compare the reviewed approaches on methodology, reported detection performance, scalability and deployment readiness.
- To identify the limitations and unresolved challenges shared across the existing body of research.
- To derive concrete research gaps and future research directions that are directly relevant to the design of the assigned platform.

1.3 Survey Methodology

Candidate papers were identified by keyword search across IEEE Xplore, ScienceDirect, MDPI journals, SpringerLink, Nature-family journals and the ACM Digital Library, using search terms including “water leak detection”, “IoT pipeline monitoring”, “SCADA leakage detection”, “digital twin water distribution network”, “machine learning leak detection” and “GIS spatial leak detection”. Candidates were screened for direct relevance to potable-water distribution networks (excluding oil/gas-only pipeline studies and non-water IoT domains), for having an accessible, verifiable methodology, and for collectively covering the sensing, analytics, control and visualization layers relevant to the assigned platform. Figure 1 summarizes the search and selection process; Figure 2 organizes the ten selected papers into a taxonomy used throughout this survey.

Figure 1: Literature Search and Selection Process*Figure 1: Literature Search and Selection Process***Figure 2: Taxonomy of Reviewed Leak-Detection Approaches**

Each branch maps to a component of the assigned platform: sensing layer, edge inference, SCADA valve/pump control, and the GIS-based monitoring dashboard.

Figure 2: Taxonomy of Reviewed Leak-Detection Approaches

2. Individual Paper Reviews

Each of the ten selected papers is reviewed below in a consistent format covering its objective, methodology, technology and dataset, and key findings, in the order shown in the taxonomy in Figure 2: broad reviews first, followed by sensing/hardware-focused studies, machine-learning comparison studies, and finally digital-twin and GIS-based methods.

2.1 Islam et al. (2022) — A Review on Current Technologies and Future Direction of Water Leakage Detection [1]

This IEEE Access review systematically analyses 47 articles on water-leakage detection and localization to surface research trends and future directions, guided by four research questions covering how the field evolved over time, which sensors are used, which techniques dominate, and where research gaps remain. The review finds that research has progressively shifted toward combining IoT connectivity with machine learning, and that acoustic, vibration and flow sensors are the most commonly used sensing modalities because they are inexpensive and straightforward to install. As a broad landscape-mapping review rather than an original technical system, it establishes the vocabulary and context against which the more specific papers below can be read.

2.2 Romero Ben et al. (2023) — Leak Detection and Localization in Water Distribution Networks: Review and Perspective [2]

Published in *Annual Reviews in Control*, this paper traces the historical evolution of leak detection and localization methods from model-based approaches, which rely on a calibrated hydraulic model of the network, toward increasingly data-driven approaches that learn detection rules directly from sensor data. It compares the two families on their practical applicability in real water utilities, highlighting that model-based methods depend on accurate and continuously updated hydraulic models while data-driven methods depend on the availability of sufficiently representative training data. The review's central contribution is this comparative perspective rather than a new detection algorithm, making it a useful frame for interpreting the more specialized studies later in this survey.

2.3 Zuñiga-Urbe et al. (2026) — Artificial Intelligence in Water Distribution Networks: A Systematic Review [3]

This PRISMA-guided systematic review in *Smart Cities* analyses 53 studies published between 2018 and 2025 that apply machine learning, deep learning or hybrid models to leak detection. It reports that pressure is the most widely used and most sensitive input variable for identifying hydraulic anomalies, and that the majority of reviewed studies rely on EPANET-simulated datasets rather than field or experimental data because collecting real operational data is costly and logistically complex. Reported detection performance in the surveyed literature ranges from 94-100% accuracy for support vector machines to 95-99% accuracy for convolutional neural networks used in multiclass classification and localization tasks, with hybrid feature-extraction-plus-classifier pipelines generally performing best. The review explicitly flags the shortage of standardized, comparable benchmark datasets as a limitation of the field as a whole.

2.4 Liu et al. (2019) — Water Pipeline Leakage Detection Based on Machine Learning and Wireless Sensor Networks [4]

Published in *Sensors*, this paper designs a leak-detection method combining a wireless sensor network (WSN) of pipeline-mounted sensors with a machine-learning classifier. Sensor data is relayed over a 4G network, and a leakage-identification method extracts a signal feature set using the intrinsic mode function, approximate entropy and principal component analysis before classifying it with a support vector machine (SVM). To address the energy cost of continuous WSN operation, the authors propose a “leakage-triggered networking” scheme that reduces energy consumption and extends node lifetime. Simulation and experimental results show the method identifies leaks effectively while consuming less energy than conventional WSN networking approaches, though validation remains at a testbed rather than municipal scale.

2.5 Boujelben et al. (2023) — An Efficient System for Water Leak Detection and Localization Based on IoT and Lightweight Deep Learning [5]

This paper in the *Internet of Things* journal targets water-scarce regions with limited infrastructure budgets. Non-invasive acoustic sensors are mounted on the pipe exterior, and a lightweight one-dimensional convolutional neural network (1D-CNN) is embedded directly on the IoT device so that only leak/no-leak classification results, not raw audio, need to be transmitted, conserving both bandwidth and energy. Detected leak information is relayed over the long-range, low-energy LoRaWAN protocol. On the authors' test rig, the classification model reached 97% accuracy, and a companion regression model localized leaks to within 0.0006 and 0.0004 coordinate units on the two spatial axes, illustrating that meaningful accuracy is achievable even on resource-constrained edge hardware.

2.6 Islam et al. (2023) — An Intelligent IoT and ML-Based Water Leakage Detection System [6]

In this follow-up IEEE Access paper, the same research group that produced the 2022 review develops its own edge machine-learning system. The device was built in three stages: first capturing real-life audio recordings of leak and non-leak conditions, then developing a classification model from that data, and finally embedding the trained model on a low-power IoT device so detection happens locally rather than in the cloud. The explicit motivation is that traditional leakage-detection methods are time-consuming and inefficient, and that onsite, real-time detection can reduce both water loss and the environmental and economic costs that follow from delayed detection.

2.7 Joseph et al. (2025) — Early Leak and Burst Detection in Water Pipeline Networks Using Machine Learning Approaches [7]

This Water (MDPI) paper conducts a comprehensive comparison of fourteen machine-learning algorithms on multivariate time-series data collected under leak, major-leak and no-leak conditions at an experimental site, framing detection as a three-class classification problem. Evaluation covers accuracy, precision, recall and F1-score using both micro- and macro-averaged metrics. Random Forest, K-Nearest Neighbours and Decision Tree emerged as the strongest performers, effectively capturing subtle patterns in the sensor data and automating what would otherwise be a manual classification task. The scale of the comparison, fourteen algorithms evaluated under a consistent experimental protocol, makes this one of the more rigorous benchmarking studies among the papers reviewed here.

2.8 Joseph et al. (2024) — Leak and Burst Detection in Water Distribution Network Using Logic- and Machine Learning-Based Approaches [8]

This companion Water (MDPI) study by an overlapping author group evaluates leak and burst detection on live, real-world infrastructure: a pumping main in the Sunbury region of Victoria, Australia, instrumented with Supervisory Control and Data Acquisition (SCADA) sensors for pressure, flow and pump speed. The authors compare simple logic-based (rule-based) algorithms against several machine-learning models, with the Local Outlier Factor model standing out for anomaly detection, and develop a web-based platform to operationalize the chosen model. Their central finding is that the two families are complementary: logic-based rules excel at straightforward, predefined anomaly patterns, while machine-learning models adapt better to complex or evolving conditions, together forming a more complete detection strategy than either alone.

2.9 Pandey et al. (2024) — Machine Learning-Based Digital Twin for Water Distribution Network Anomaly Detection and Localization [9]

Presented at the WDSA/CCWI 2024 conference and published in Engineering Proceedings, this paper builds a digital twin of a water distribution network for leak detection and localization using two data-driven model designs. A single-stage model applies Logistic Regression and Random Forest directly to flag anomalies, while a two-stage model first uses linear regression to predict expected pressure differences between sensor pairs and then classifies deviations in the resulting residual distribution to estimate the posterior probability of a leak at each candidate location. Both approaches were tested on a real-time operational dataset, demonstrating that a digital twin can support not just anomaly flagging but probabilistic leak localization, though the paper stops at detection and localization rather than closing the loop with automated control action.

2.10 Elshazly et al. (2024) — An Automated GIS-Based Spatial Machine Learning Method for Leak Detection [10]

Published in Applied Sciences, this paper proposes a leak-detection method that deliberately reuses the pressure, flow and water-quality monitoring sensors a utility has already installed, rather than requiring a new dedicated sensor network. It combines Geographically Weighted Regression (GWR), which models how the relationship between variables changes across geographic space, with the Local Outlier Factor (LOF) technique to flag spatial anomalies, applied to a real water distribution network dataset supplied by a partner utility authority. Because the method is explicitly map-integrated, it connects naturally to a GIS-based operations dashboard of the kind the assigned platform is intended to provide, at the cost of being bounded by the density and placement of whatever sensors the utility already has installed.

3. Comparative Analysis

3.1 Comparative Literature Survey Table

Table 1 consolidates the ten reviewed papers against the parameters required for this exercise: the research problem addressed, methodology used, dataset/tools, key findings, limitations, and the specific research gap each paper leaves open.

Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
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Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
Islam et al., 2022 [1]	Map the landscape of water-leakage detection research	Systematic literature review of 47 articles; 4 guiding research questions	Web of Science / IEEE / Scopus literature corpus	Research trending toward ML+IoT convergence; acoustic, vibration and flow sensors dominate (low cost, easy install)	Review scope limited to articles indexed up to 2022; fast-moving field ages quickly	Lack of consolidated, comparable benchmarking across independently reported studies
Romero Ben et al., 2023 [2]	Evolution and practical applicability of leak localization methods	Narrative review comparing model-based (hydraulic) vs. data-driven method families	Prior published methods across both families	Traces a shift from model-based to data-driven approaches; discusses real-utility applicability trade-offs	Model-based methods need costly, well-calibrated hydraulic models; data-driven methods need large representative training sets	Little work unifying both families into a single adaptive, utility-deployable framework
Zuñiga-Uribe et al., 2026 [3]	Consolidate ML/DL leak-detection model performance and inputs	PRISMA-guided systematic review of 53 studies (2018–2025)	EPANET-simulated data (majority); limited field/experimental data	Pressure is the most sensitive input variable; SVMs reach 94–100% accuracy, CNNs 95–99% for multiclass/localization	Most reported accuracies come from simulated rather than field data, limiting real-world transfer	Scarcity of standardized, real-world benchmark datasets for cross-study comparison
Liu et al., 2019 [4]	Energy-efficient leak detection over a wireless sensor network	WSN of pipeline-mounted sensors + 4G backhaul; IMF + approximate entropy + PCA features; SVM classifier; leakage-triggered networking	Simulation plus a physical pipeline testbed	Leakage-triggered networking lowers WSN energy consumption versus conventional networking while identifying leaks effectively	Feature-engineering pipeline may not generalize across pipe materials/diameters; validated at limited scale, not municipal deployment	Need for models that generalize across pipe types with less manual feature engineering
Boujelben et al., 2023 [5]	Low-power, real-time leak detection and localization for resource-constrained regions	Non-invasive acoustic sensors + on-device lightweight 1D-CNN (classification and localization) + LoRaWAN uplink	Acoustic signals from an experimental pipe/pump testbed	97% leak-classification accuracy; localization error of 0.0006 (X) and 0.0004 (Y) coordinate units on the testbed	Evaluated on a local/limited acoustic dataset and a specific testbed geometry; LoRaWAN's low data rate constrains payload richness	Field-scale, multi-zone validation and generalization across pipe materials and ambient-noise conditions
Islam et al., 2023 [6]	On-device (edge) leak detection to avoid streaming raw sensor data	Edge ML pipeline: controlled leak/non-leak audio capture, model development, and on-device embedding on a low-power IoT device	Real-life audio data captured by the authors' own experimental setup	Demonstrates a working low-power edge-ML device that classifies leaks locally without continuous cloud transmission	Acoustic-only sensing may degrade in noisy urban environments; low-power budget limits sensing frequency	Multimodal fusion (acoustic + pressure/flow) for robustness where acoustic signal alone is unreliable
Joseph et al., 2025 [7]	Compare lightweight ML algorithms for multi-class leak/burst classification	Comparison of 14 ML algorithms on IoT-derived multivariate time-series data; 3-class classification (no-leak/leak/burst)	Data from an experimental site under controlled leak, major-leak and no-leak conditions	Random Forest, K-Nearest Neighbours and Decision Tree outperform other algorithms on accuracy, precision, recall and F1-score	Single experimental-site dataset, not multi-utility field data; focuses on classical/lightweight ML rather than deep sequence models	Benchmarking across heterogeneous, larger real distribution networks and against deep sequence models

Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
Joseph et al., 2024 [8]	Real-world leak/burst detection on live SCADA infrastructure	Comparison of logic (rule)-based algorithms and ML models (incl. Local Outlier Factor) on live SCADA pressure/flow/pump-speed data; web-based deployment platform	Live SCADA data from a pumping main in Sunbury, Victoria, Australia	Logic-based and ML approaches are complementary: logic excels at simple predefined anomalies, ML adapts to complex/evolving patterns; LOF stood out among ML models	Single pumping-main case study; rule thresholds require manual per-site tuning; false-alarm risk noted by the authors	Scaling the hybrid logic+ML approach across many heterogeneous DMAs without manual retuning; quantifying alert-fatigue impact
Pandey et al., 2024 [9]	Detect and localize leaks using a digital twin of the network	Single-stage (Logistic Regression/Random Forest anomaly test) and two-stage (residual-based posterior probability) data-driven digital-twin models	A real-time operational dataset from a water distribution network	Two-stage residual model computes posterior probabilities of leak location from inter-sensor pressure-difference patterns	Conference-length study; digital twin covers detection/localization only, not closed-loop control actuation	End-to-end closed-loop digital twins connecting detection directly to automated or human-approved valve/pump actuation
Elshazly et al., 2024 [10]	Reuse existing utility monitoring sensors for spatial leak detection	GIS-based spatial ML using Geographically Weighted Regression (GWR) and Local Outlier Factor (LOF)	Real WDN pressure/flow/quality data supplied by a partner utility authority	Automated, map-integrated leak flagging built directly on infrastructure utilities already operate, without new dedicated sensor networks	Detection resolution is bounded by existing sensor density/placement; GWR/LOF are comparatively simple versus deep spatio-temporal models	Combining GIS spatial analytics with denser IoT retrofits and more expressive models (e.g., graph neural networks) for finer localization

3.2 Comparison by Detection Performance

Where papers report a headline performance figure, accuracy clusters in a high but not uniform band: the systematic review by Zuñiga-Urbe et al. [3] reports 94-100% accuracy for SVM-based studies and 95-99% for CNN-based studies across the literature it surveys; Boujelben et al. [5] report 97% leak-classification accuracy for their own lightweight 1D-CNN; and Joseph et al. [7] find Random Forest, KNN and Decision Tree superior among fourteen compared algorithms without a single consolidated percentage, since performance is reported per metric (accuracy, precision, recall, F1) and per class (no-leak, minor leak, major leak). Other papers, including the WSN/SVM study by Liu et al. [4], the SCADA case study by Joseph et al. [8], the digital twin by Pandey et al. [9], and the GIS-spatial method by Elshazly et al. [10], report qualitative superiority over baseline or rule-based methods rather than a single benchmark figure, which itself illustrates the benchmarking inconsistency flagged in Section 3.1 and revisited in Section 4.

3.3 Comparison by Methodology Type

The reviewed literature falls into five broad methodology families, mirrored in Figure 2. Reviews and systematic reviews [1], [2], [3] synthesize the field rather than proposing new systems. WSN-plus-classical-ML approaches [4] rely on hand-engineered signal features feeding a conventional classifier such as an SVM. Edge and deep-learning acoustic approaches [5], [6] push inference onto low-power devices using lightweight CNNs, trading some model capacity for bandwidth and energy savings. SCADA-based ML comparison studies [7], [8] evaluate multiple algorithms, including rule-based logic, on operational or near-operational data. Digital-twin and GIS-spatial methods [9], [10] add a model of the network's physical or geographic structure on top of the raw sensor stream, enabling localization rather than detection alone.

3.4 Comparison by Scalability and Deployment Readiness

Deployment readiness varies considerably. The SCADA case study [8] and the GIS-spatial method [10] were both evaluated against real utility infrastructure and existing sensor deployments, giving them a head start toward production use. The edge-ML papers [5], [6] target low-power, low-bandwidth hardware explicitly for wide, cost-constrained rollout, but validation remains at testbed scale. The WSN paper [4] addresses energy efficiency, a prerequisite for scaling to a large sensor count, but again at testbed scale. The digital-twin paper [9] demonstrates a technique that could scale to a full network if paired with a sufficiently detailed hydraulic model, but does not itself report a large-scale deployment. None of the ten papers report a live, multi-DMA, utility-wide rollout that also includes automated valve or pump control, which is precisely the gap the assigned platform's design must address.

4. Limitations and Challenges of Existing Approaches

Beyond the paper-specific limitations listed in Table 1, several challenges recur across the body of literature as a whole:

Data and benchmarking limitations

As Zuñiga-Urbe et al. [3] make explicit, most machine-learning and deep-learning studies in this field train and evaluate on EPANET-simulated data rather than field data, because collecting labeled real-world leak events is expensive and operationally disruptive. This means reported accuracy figures are not always directly comparable across studies, and high simulated-data accuracy does not guarantee equivalent performance on noisy, real infrastructure.

Field validation and generalization

Several of the more technically detailed papers [4], [5], [9] validate their methods on a single testbed, laboratory rig, or limited real-time dataset rather than across multiple, heterogeneous distribution networks. Only Joseph et al. [8] and Elshazly et al. [10] evaluate against live, real-world utility infrastructure, and each does so at a single site.

Sensor and hardware constraints

Edge-deployed models [5], [6] must balance detection accuracy against the power, memory and bandwidth budgets of low-cost IoT hardware, and acoustic sensing in particular can degrade in noisy environments. WSN-based approaches [4] must additionally manage network energy consumption at scale, which becomes harder as sensor count grows.

Integration with control and legacy systems

Only the SCADA case study [8] works directly with an operational control system, and even there the paper addresses detection rather than automated valve or pump actuation in response to a confirmed leak. None of the reviewed papers describe a closed-loop system that both detects a leak and safely triggers a physical control action.

Explainability and operator trust

With the exception of the model-versus-logic comparison in [8], the reviewed papers evaluate models primarily on statistical accuracy rather than on whether a utility operator can understand and trust a given alert enough to act on it, an issue that becomes more important as models move from academic benchmarks toward systems that can trigger real infrastructure changes.

False positives and alert fatigue

Few papers quantify false-positive rate as an operational concern in its own right, even though an excessive rate of false alerts is what erodes operator confidence in an automated system in practice, as implicitly acknowledged by the logic-vs-ML comparison in [8].

5. Identified Research Gaps

Synthesizing Sections 3 and 4, the following specific, unresolved gaps emerge from the reviewed literature:

Gap	Description
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Gap	Description
RG1	No reviewed paper integrates leak detection, SCADA-based valve/pump isolation, and maintenance work-order dispatch into a single closed-loop, field-validated system; most work stops at detection and localization.
RG2	Standardized, publicly available, multi-utility real-world benchmark datasets remain scarce; the majority of reported results rely on EPANET simulation or single-site data, limiting reliable cross-study comparison [3].
RG3	Multimodal sensor fusion (acoustic + pressure/flow + spatial/GIS context) is rarely combined within one deployed system; each reviewed paper specializes in a single sensing modality.
RG4	Explainability and operator trust are largely unaddressed; models are evaluated on accuracy alone rather than on how confidently a human operator can act on an alert before approving a control action.
RG5	Digital twins are validated for detection and localization [9] but not yet closed with real actuation; the safety implications of granting an automated model authority over physical valve/pump commands remain underexplored.
RG6	False-positive rate and alert fatigue are rarely reported as first-class operational metrics, despite their direct effect on utility staff workload and on trust in an automated system [8].

6. Future Research Directions

Each proposed direction below responds to one of the research gaps identified in Section 5, and each is chosen for direct relevance to the assigned Intelligent Water Distribution Monitoring and Leakage Detection Platform.

Direction	Description
FD1 (addresses RG1)	Extend detection/localization research into closed-loop pipelines that couple a confirmed leak alert directly to automated (or human-approved) valve isolation and an auto-generated work order — the leak-to-repair workflow the assigned platform is designed around.
FD2 (addresses RG2)	Publish a field-validated benchmark dataset from a real multi-DMA deployment (flow, pressure and acoustic telemetry with labeled leak events); the assigned platform's staging environment and digital-twin sensor simulator are well positioned to generate such a dataset during rollout.
FD3 (addresses RG3)	Investigate multimodal fusion models that combine acoustic, pressure/flow and GIS-spatial features, since each single-modality system reviewed here showed characteristic blind spots that another modality could plausibly cover.
FD4 (addresses RG4)	Apply explainable-AI techniques (e.g., feature-attribution methods or interpretable graph-based models) to leak alerts so operators can quickly validate a model's reasoning before confirming a leak or issuing a control command.
FD5 (addresses RG5)	Study staged, shadow-mode rollout strategies for control-layer model updates, in which a new detection model's recommendations run in parallel with the production model before being trusted with authority over real valve or pump actuation.
FD6 (addresses RG6)	Track and optimize false-positive rate and alert fatigue as explicit, first-class deployment metrics alongside detection accuracy, rather than reporting accuracy alone.

7. Conclusion

This survey reviewed ten peer-reviewed papers spanning broad literature reviews, wireless-sensor-network and classical machine-learning methods, edge and deep-learning acoustic sensing, SCADA-based comparison studies on live infrastructure, and digital-twin and GIS-spatial approaches to water-pipeline leak detection. Across this body of work, sensing and detection methods are comparatively mature: pressure and acoustic sensing combined with machine learning or lightweight deep learning routinely achieves high reported accuracy, and at least one study validates its approach against live utility SCADA data. What remains underdeveloped is the layer above detection: closed-loop integration with automated control, standardized field benchmarking, multimodal fusion, explainability for operator trust, and false-positive management as an operational rather than purely statistical concern.

These gaps map directly onto the design of the assigned Intelligent Water Distribution Monitoring and Leakage Detection Platform, which combines IoT sensing, a leak-detection algorithm, a GIS-based dashboard, SCADA-integrated valve/pump control, and automated work-order dispatch in a single system. The future directions proposed in Section 6, particularly closed-loop detection-to-actuation workflows, field-generated benchmark datasets, and staged/shadow-mode rollout of control-layer model updates, are offered as concrete, literature-grounded next steps for that platform's continued development.

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